```
Start coding or generate with AI.
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("debajyotipodder/co2-emission-by-vehicles")

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/debajyotipodder/co2-emission-by-vehicles/1.archive.
100%|██████████| 88.3k/88.3k [00:00<00:00, 11.3MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/debajyotipodder/co2-emission-by-vehicles/ve
```

Once the dataset is downloaded, we need to load it into a pandas DataFrame. Typically, Kaggle datasets contain CSV files. Let's assume the main data file is a CSV and try to find it within the downloaded path.

```
import pandas as pd
import os

# Find the CSV file in the downloaded path
csv_files = [f for f in os.listdir(path) if f.endswith('.csv')]

if csv_files:
    # Assuming the first CSV file is the main dataset
    data_file = os.path.join(path, csv_files[0])
    df = pd.read_csv(data_file)
    print(f"Loaded data from: {data_file}")
    display(df.head())
else:
    print("No CSV files found in the downloaded dataset path.")
    df = None # Set df to None if no CSV found
```

```
Loaded data from: /root/.cache/kagglehub/datasets/debajyotipodder/co2-emission-by-vehicles/version
```

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Co Co
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9	6.7	
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2	7.7	
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0	5.8	
3	ACURA	MDX 4WD	SUV - SMALL	3.5	6	AS6	Z	12.7	9.1	
4	ACURA	RDX AWD	SUV - SMALL	3.5	6	AS6	Z	12.1	8.7	

Now that we have loaded the data, let's proceed with the steps you outlined:

1. **Select features like Engine Size, Cylinders, Fuel Consumption.**
2. **Target variable: CO2 Emissions.**

3. **Implement Multiple Linear Regression.**
4. **Evaluate model using RMSE and R^2 score.**
5. **Interpret coefficients and explain environmental impact.**

```
# Step 1 & 2: Select features and target variable

# Let's inspect the column names to make sure we use the correct ones.
# This is a common practice before selecting columns definitively.
print("Available columns:", df.columns.tolist())

# Based on common dataset names, let's assume:
# Engine Size -> 'Engine Size(L)' or similar
# Cylinders -> 'Cylinders'
# Fuel Consumption -> 'Fuel Consumption City (L/100 km)', 'Fuel Consumption Hwy (L/100 km)', 'Fue
# CO2 Emissions -> 'CO2 Emissions(g/km)' or similar

# We need to map the user's requested names to the actual column names in the dataframe.
# For now, I will use common possible column names. If they are incorrect, we can adjust after se

# Example assumed column names:
feature_cols = ['Engine Size(L)', 'Cylinders', 'Fuel Consumption Comb (L/100 km)']
target_col = 'CO2 Emissions(g/km)'

# Check if the assumed columns exist in the DataFrame
missing_features = [col for col in feature_cols if col not in df.columns]
missing_target = target_col not in df.columns

if missing_features or missing_target:
    print("Warning: Some expected columns are missing. Please verify column names from df.columns
    if missing_features:
        print(f"Missing features: {missing_features}")
    if missing_target:
        print(f"Missing target: {target_col}")
else:
    X = df[feature_cols]
    y = df[target_col]

    print(f"Features selected: {feature_cols}")
    print(f"Target variable: {target_col}")
    print("\nFirst 5 rows of features (X):")
    display(X.head())
    print("\nFirst 5 rows of target (y):")
    display(y.head())
```

```
Available columns: ['Make', 'Model', 'Vehicle Class', 'Engine Size(L)', 'Cylinders', 'Transmission
Features selected: ['Engine Size(L)', 'Cylinders', 'Fuel Consumption Comb (L/100 km)']
Target variable: CO2 Emissions(g/km)

First 5 rows of features (X):
```

	Engine Size(L)	Cylinders	Fuel Consumption Comb (L/100 km)
0	2.0	4	8.5
1	2.4	4	9.6
2	1.5	4	5.9
3	3.5	6	11.1
4	3.5	6	10.6

```
First 5 rows of target (y):
```

	CO2 Emissions(g/km)
0	196
1	221
2	136
3	255
4	244

dtype: int64

```
# Step 3: Implement Multiple Linear Regression
# Step 4: Evaluate model using RMSE and R² score

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

if df is not None and not missing_features and not missing_target:
    # Split the data into training and testing sets
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    # Initialize and train the Linear Regression model
    model = LinearRegression()
    model.fit(X_train, y_train)

    # Make predictions on the test set
    y_pred = model.predict(X_test)

    # Evaluate the model
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)

    print(f"Model Training Complete.")
    print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")
    print(f"R-squared (R²): {r2:.2f}")
else:
    print("Cannot perform regression. Please ensure data is loaded and columns are correctly ider
```

```
Model Training Complete.
Root Mean Squared Error (RMSE): 20.54
R-squared (R²): 0.88
```

```
# Step 5: Interpret coefficients and explain environmental impact

if df is not None and not missing_features and not missing_target:
    print("Model Coefficients:")
    for i, col in enumerate(feature_cols):
        print(f"{col}: {model.coef_[i]:.2f}")
    print(f"Intercept: {model.intercept_:.2f}")

    print("\nEnvironmental Impact Interpretation:")
    print("The coefficients indicate the change in CO2 Emissions (g/km) for a one-unit increase i
    print("For example:")
    print(f"- A {model.coef_[feature_cols.index('Engine Size(L)')]:.2f} increase in CO2 emissions
    print(f"- A {model.coef_[feature_cols.index('Cylinders')]:.2f} increase in CO2 emissions for
    print(f"- A {model.coef_[feature_cols.index('Fuel Consumption Comb (L/100 km)')]:.2f} increas
    print("\nGenerally, positive coefficients indicate that an increase in that feature leads to
    print("These insights can help in designing more fuel-efficient vehicles and promoting sustai
else:
    print("Cannot interpret coefficients. Please ensure data is loaded and model is trained.")
```

```
Model Coefficients:
Engine Size(L): 5.59
Cylinders: 6.38
Fuel Consumption Comb (L/100 km): 13.27
Intercept: 51.53

Environmental Impact Interpretation:
The coefficients indicate the change in CO2 Emissions (g/km) for a one-unit increase in the corres
For example:
- A 5.59 increase in CO2 emissions for every 1-liter increase in Engine Size.
- A 6.38 increase in CO2 emissions for every additional Cylinder.
- A 13.27 increase in CO2 emissions for every 1 L/100 km increase in Combined Fuel Consumption.

Generally, positive coefficients indicate that an increase in that feature leads to higher CO2 emi
These insights can help in designing more fuel-efficient vehicles and promoting sustainable transp
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